MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
3610 POX CONDITIONING, LEACHING & ADSORPTION
FOUNDATION FOR REA- 001 & 002

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MD-216-3610-EG-CV-CAL-1001	**Mansourah Massarah Gold Project**	
900707_GAU01_HL12_CED01_ 36103_OU100680999_R2	**DESIGN CALCULATION POX CONDITIONING, LEACHING & ADSORPTION FOUNDATION FOR REA – 001 & 002**	**Date: 22.11.2023**
O19045_450_C_K_24632_001_R2	**AS BUILT**	**Rev: D01**

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is to Prepare for design of **Pox Conditioning, leaching & Adsorption (3610) REA-001 & 002-** Tanks' Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel. For member strength calculation force values are directly taken from Load Data Provided.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Foundation and Load Plan (Doc. No. MD-216-3610-EG-ME-PLN-1001) attached in Appendix A. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5 CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundations for Aboveground Tanks

Foundation shall be Earth Foundation with a Concrete Ringwall. The design calculations are prepared as per recommendation of standard API 650 - Annex B.

5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Wall	50

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5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m^2 at 1.2m depth below EGL (ie. RL 911.50m) refer Table 5 - BH-11 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

6. DESIGN CALCULATION FOR FOUNDATION

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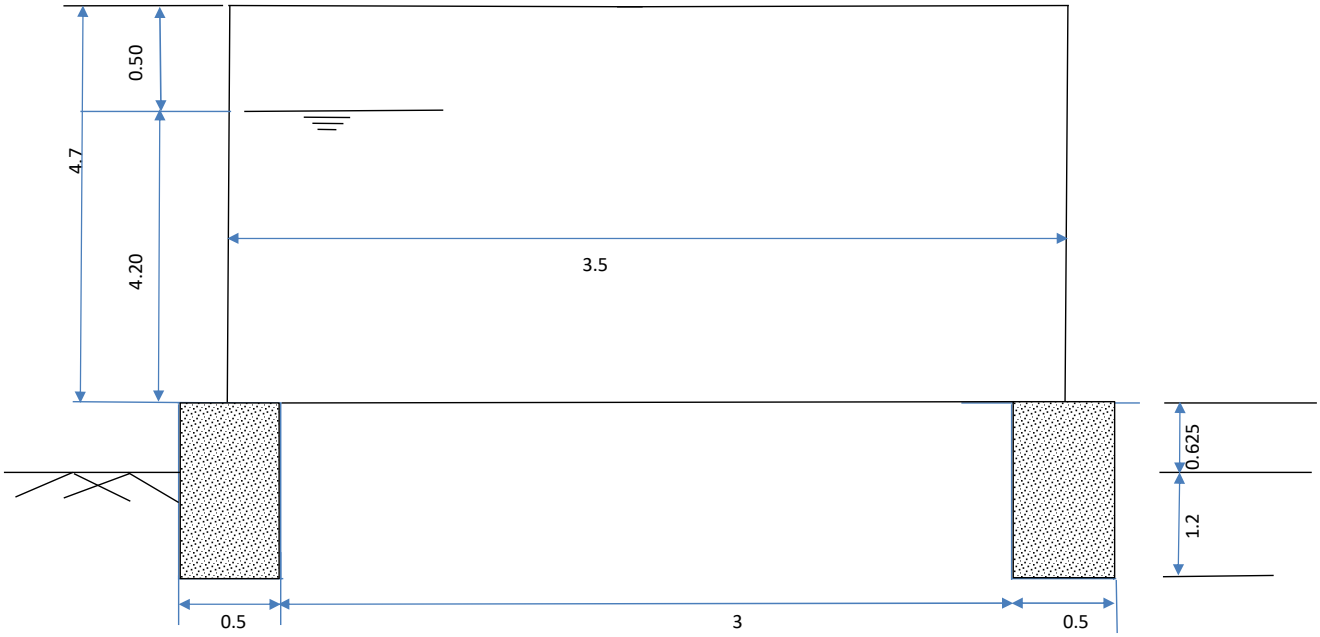
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Foundation Marked F1

Tank Loading Data

DEAD WEIGHT OF EQUIPMENT	74	kN
DEAD WEIGHT OF CIVIL STRUCTURES	0	kN
OPERATIONAL WEIGHT OF SLURRY	529	kN
MAXIMUM WEIGHT OF SLURRY	592	kN
EXTERNAL PLATFORM LOAD	17.6	kN
WIND LOAD	17	kN
SEISMIC LOAD	26.1	kN
SEISMIC TORQUE AT FOUNDATION LEVEL	120	kNm

	Loading Type		
Weight, V	Empty (W)	91.6	kN
	Operating (Wo)	620.6	kN
	Test (Wt)	683.6	kN



Diameter of Tank, D	=	3.5 m	
Height of Tank, H	=	4.7 m	
Height of Tank Fluid, H_L	=	4.2 m	
Inner Diameter of Concrete Ring Wall, ID	=	3 m	
Outer Diameter of Concrete Ring Wall, OD	=	4 m	
Height of Concrete Ring Wall, h	=	1.825 m	
ThickNess of Concrete Ring Wall, t	=	0.5 m	
B.C.D	=	3.75 m	
Bottom of Grout	=	0.625 m	(Above Finished Level)
Unit Weight of Liquid, γ_L	=	13.1 kN/m3	
Unit Weight of Soil, γ_s	=	18 kN/m3	
Unit Weight of Concrete, γ_c	=	24 kN/m3	

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Wind Load & Seismic Load

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Empty	17.00	53.55
	Operating	17.00	53.55
Seismic Loads	Operating	26.10	120.00

Foundation Weight

a) Ring Wall Weight = 240.80 kN

a) Inside Soil Weight = 232.20 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 5.50 m^2

Full cross-sectional area for operating condition, Ao= 12.57 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 4.30 m^3

Sectional modulus for operating condition,Zo = 6.28 m^3

Check of Minimum Wall ThickNess (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to tank,minimum thickNess of concrete wall required to equalize soil pressures between under tank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma c\ h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma c\ h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 8.33 kN/m

So, Required Min Wall THK.= 0.5031 m = 503.1 mm > 500 mm OK

Basic Load Cases

1. Tank Dead Load (D)
2. Operational Fluid Load (F1)
3. Test/Max Fluid Load (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 ,2007 Cl No. 2.4.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101	D + F1	620.60	0.00	0.00
Short term	Empty	LC102	0.6 D + W	54.96	17.00	53.55
	Operating	LC103	D + F1 + W	620.60	17.00	53.55
		LC104	D + F1+ 0.7E	620.60	18.27	84.00
		LC105	0.6 (D + F1) +0.7 E	372.36	18.27	84.00
	Test	LC106	D + F2	683.60	0.00	0.00

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 450 kN/m2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 598.50 kN/m2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m2	Qmax= V/A - M/Z kN/m2	Qall/ Qmax	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	1093.61	0.00	87.03	87.03	5.17	OK
LC102	295.76	84.58	73.49	34.11	8.14	OK
LC103	1093.61	84.58	100.49	73.57	5.96	OK
LC104	1093.61	117.34	105.70	68.35	5.66	OK
LC105	845.37	117.34	85.95	48.60	6.96	OK
LC106	1156.61	0.00	92.04	92.04	4.89	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of foundation level	Coef. of friction	Hresist (kN)	Hacting (kN)	Fs	Sliding Check
	V (kN)					
LC101	1093.61	0.4	437.44	0.00	-	-
LC102	295.76	0.4	118.31	17.00	7.0	OK
LC103	1093.61	0.4	437.44	17.00	25.7	OK
LC104	1093.61	0.4	437.44	18.27	23.9	OK
LC105	845.37	0.4	338.15	18.27	18.5	OK
LC106	1156.61	0.4	462.64	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of foundation level	Radius of fdn.	Mresist (kNm)	Macting (kNm)	Fs	Overturning Check
	V (kN)					
LC101	1093.61	2	2187.21	0.00	-	-
LC102	295.76	2	591.53	84.58	7.0	OK
LC103	1093.61	2	2187.21	84.58	25.9	OK
LC104	1093.61	2	2187.21	117.34	18.6	OK
LC105	845.37	2	1690.73	117.34	14.4	OK
LC106	1156.61	2	2313.21	0.00	-	-

Ultimate Strength Design

Load Combination (As per SBC-304 ,2018 Table 5.3.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	868.84	0.00	9.62	4.21
Short term	Empty	LC102	0.9 D + 1.6W	82.44	53.55	9.62	4.21
	Operating	LC103	1.2 (D + F1) +1.6 W	744.72	53.55	9.62	4.21
		LC104	1.2 (D + F1) + E	744.72	120.00	9.62	4.21
		LC105	0.9 (D + F1) + E	558.54	120.00	9.62	4.21
	Test	LC106	1.4 (D + F2)	957.04	0.00	9.62	4.21

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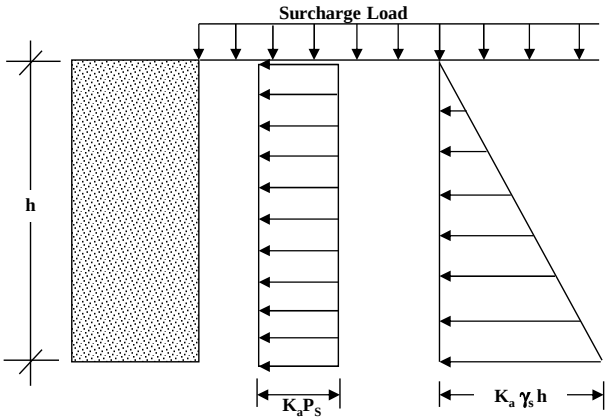
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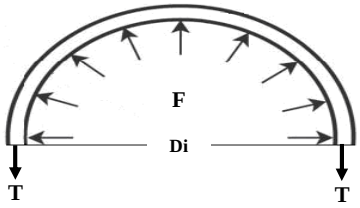
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Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m2
LC101	868.84	0.00	90.31
LC102	82.44	53.55	21.29
LC103	744.72	53.55	90.13
LC104	744.72	120.00	105.91
LC105	558.54	120.00	86.56
LC106	957.04	0.00	99.47

Maximum Surcharge Load, Ps = 105.91 kN/m2





Where, K_a = coefficient of earth pressure at rest
= 0.5

P1 = $K_a * P_s$ = 52.96 kN/m2

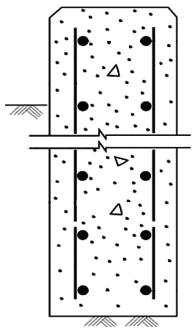
P2 = $1.4 * \gamma_S * h * K_a$ = 23.00 kN/m2 for LC106 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 117.629 kN/m

Hoop Tension Load

T= 1/2 * F * ID = 176.443 kN



Hoop Tension Bar

1. Horizontal reinforcement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(φ x Fy) = 2281.25 mm^2

Minimum As = 0.0025 x b x D = 2281.25 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 16 Nos. D16= 2903.334 mm^2 OK

2. Vertical reinforcement

Minimum As = 0.0015 * b * D = 750 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row * 5.00 ea - D10 (@ 200mm)

=785mm2 > Required As OK

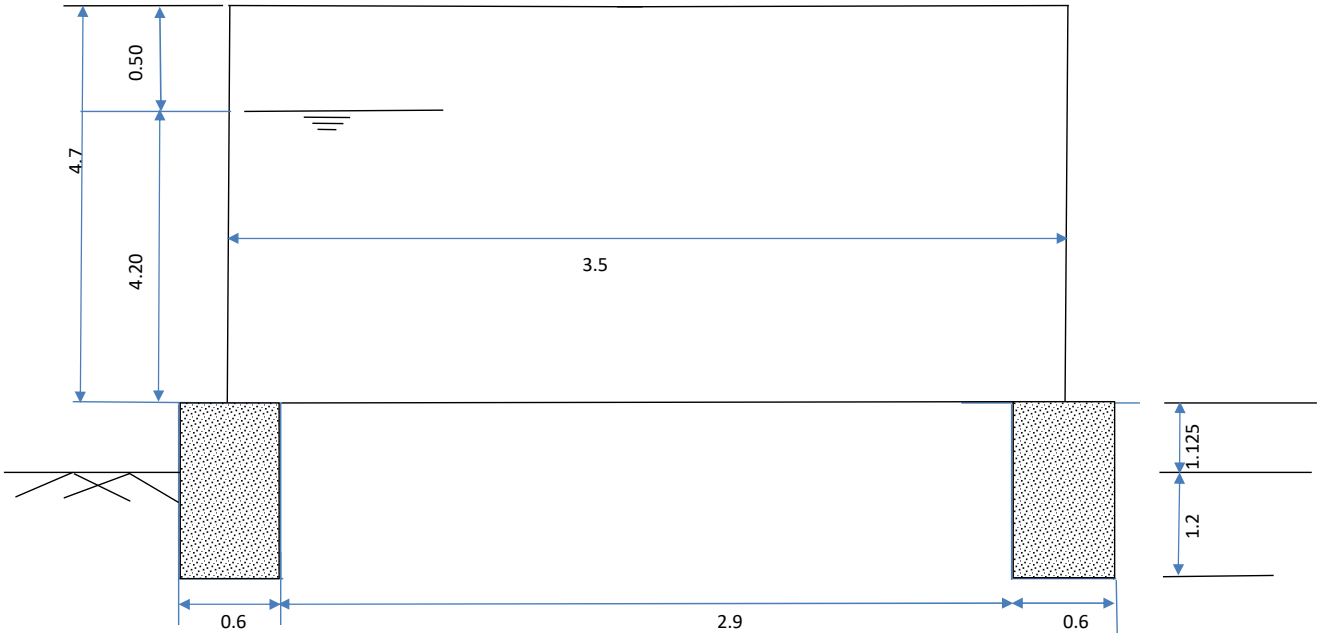
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Foundation Marked F2

Tank Loading Data

DEAD WEIGHT OF EQUIPMENT	74	kN
DEAD WEIGHT OF CIVIL STRUCTURES	0	kN
OPERATIONAL WEIGHT OF SLURRY	529	kN
MAXIMUM WEIGHT OF SLURRY	592	kN
EXTERNAL PLATFORM LOAD	17.6	kN
WIND LOAD	17	kN
SEISMIC LOAD	26.1	kN
SEISMIC TORQUE AT FOUNDATION LEVEL	120	kNm

	Loading Type		
Weight, V	Empty (W)	91.6	kN
	Operating (Wo)	620.6	kN
	Test (Wt)	683.6	kN



Diameter of Tank, D	=	3.5 m	
Height of Tank, H	=	4.7 m	
Height of Tank Fluid, H_L	=	4.2 m	
Inner Diameter of Concrete Ring Wall, ID	=	2.9 m	
Outer Diameter of Concrete Ring Wall, OD	=	4.1 m	
Height of Concrete Ring Wall, h	=	2.325 m	
ThickNess of Concrete Ring Wall, t	=	0.6 m	
B.C.D	=	3.75 m	
Bottom of Grout	=	1.125 m	(Above Finished Level)
Unit Weight of Liquid, γ_L	=	13.1 kN/m3	
Unit Weight of Soil, γ_s	=	18 kN/m3	
Unit Weight of Concrete, γ_c	=	24 kN/m3	

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Wind Load & Seismic Load

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Empty	17.00	53.55
	Operating	17.00	53.55
Seismic Loads	Operating	26.10	120.00

Foundation Weight

a) Ring Wall Weight = 368.13 kN

a) Inside Soil Weight = 276.43 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 6.60 m^2

Full cross-sectional area for operating condition, Ao= 13.20 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 5.07 m^3

Sectional modulus for operating condition,Zo = 6.77 m^3

Check of Minimum Wall ThickNess (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to tank,minimum thickNess of concrete wall required to equalize soil pressures between under tank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma c\ h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma c\ h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 8.33 kN/m

So, Required Min Wall THK.= 0.6144 m = 614.4 mm > 600 mm OK

Basic Load Cases

1. Tank Dead Load (D)
2. Operational Fluid Load (F1)
3. Test/Max Fluid Load (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 ,2007 Cl No. 2.4.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101	D + F1	620.60	0.00	0.00
Short term	Empty	LC102	0.6 D + W	54.96	17.00	53.55
	Operating	LC103	D + F1 + W	620.60	17.00	53.55
		LC104	D + F1+ 0.7E	620.60	18.27	84.00
		LC105	0.6 (D + F1) +0.7 E	372.36	18.27	84.00
	Test	LC106	D + F2	683.60	0.00	0.00

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900707_GAU01_HL12_CED01_ 36103_OU100680999_R2	DESIGN CALCULATION POX CONDITIONING, LEACHING & ADSORPTION FOUNDATION FOR REA - 001 & 002	Date: 22.11.2023
O19045_450_C_K_24632_001_R2	AS BUILT	Rev: D01

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 450 kN/m2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 598.50 kN/m2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m2	Qmax= V/A - M/Z kN/m2	Qall/ Qmax	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	1265.16	0.00	95.83	95.83	4.70	OK
LC102	423.09	93.08	82.48	45.78	7.26	OK
LC103	1265.16	93.08	109.58	82.07	5.46	OK
LC104	1265.16	126.48	114.52	77.13	5.23	OK
LC105	1016.92	126.48	95.72	58.33	6.25	OK
LC106	1328.16	0.00	100.60	100.60	4.47	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of foundation level	Coef. of friction	Hresist (kN)	Hacting (kN)	Fs	Sliding Check
	V (kN)					
LC101	1265.16	0.4	506.06	0.00	-	-
LC102	423.09	0.4	169.24	17.00	10.0	OK
LC103	1265.16	0.4	506.06	17.00	29.8	OK
LC104	1265.16	0.4	506.06	18.27	27.7	OK
LC105	1016.92	0.4	406.77	18.27	22.3	OK
LC106	1328.16	0.4	531.26	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of foundation level	Radius of fdn.	Mresist (kNm)	Macting (kNm)	Fs	Overturning Check
	V (kN)					
LC101	1265.16	2.05	2593.58	0.00	-	-
LC102	423.09	2.05	867.34	93.08	9.3	OK
LC103	1265.16	2.05	2593.58	93.08	27.9	OK
LC104	1265.16	2.05	2593.58	126.48	20.5	OK
LC105	1016.92	2.05	2084.68	126.48	16.5	OK
LC106	1328.16	2.05	2722.73	0.00	-	-

Ultimate Strength Design

Load Combination (As per SBC-304 ,2018 Table 5.3.1)

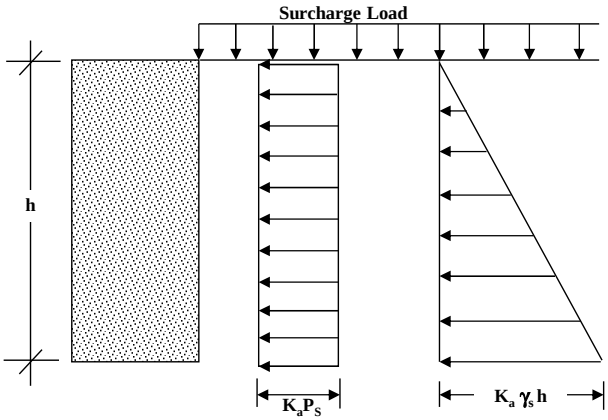
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	868.84	0.00	9.62	4.21
Short term	Empty	LC102	0.9 D + 1.6W	82.44	53.55	9.62	4.21
	Operating	LC103	1.2 (D + F1) +1.6 W	744.72	53.55	9.62	4.21
		LC104	1.2 (D + F1) + E	744.72	120.00	9.62	4.21
		LC105	0.9 (D + F1) + E	558.54	120.00	9.62	4.21
	Test	LC106	1.4 (D + F2)	957.04	0.00	9.62	4.21

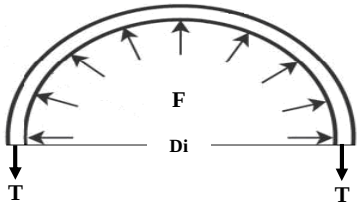
MA'ADEN معادن	Outotec / LARSEN & TOUBRO
MD-216-3610-EG-CV-CAL-1001	Mansourah Massarah Gold Project	
900707_GAU01_HL12_CED01_ 36103_OU100680999_R2	DESIGN CALCULATION POX CONDITIONING, LEACHING & ADSORPTION FOUNDATION FOR REA - 001 & 002	Date: 22.11.2023
O19045_450_C_K_24632_001_R2	AS BUILT	Rev: D01

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m2
LC101	868.84	0.00	90.31
LC102	82.44	53.55	21.29
LC103	744.72	53.55	90.13
LC104	744.72	120.00	105.91
LC105	558.54	120.00	86.56
LC106	957.04	0.00	99.47

Maximum Surcharge Load, Ps = 105.91 kN/m2





Where, K_a = coefficient of earth pressure at rest
= 0.5

P1 = $K_a * P_s$ = 52.96 kN/m2

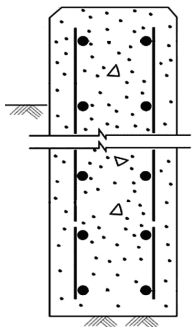
P2 = $1.4 * \gamma_S * h * K_a$ = 29.30 kN/m2 for LC106 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 157.18 kN/m

Hoop Tension Load

T= 1/2 * F * ID = 227.911 kN



Hoop Tension Bar

1. Horizontal reinforcement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(φ x Fy) = 3487.5 mm^2

Minimum As = 0.0025 x b x D = 3487.5 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 20 Nos. D16= 3629.168 mm^2 OK

2. Vertical reinforcement

Minimum As = 0.0015 * b * D = 900 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row * 6.67 ea - D10 (@ 150mm)
=1047mm2 > Required As OK

7. Reference

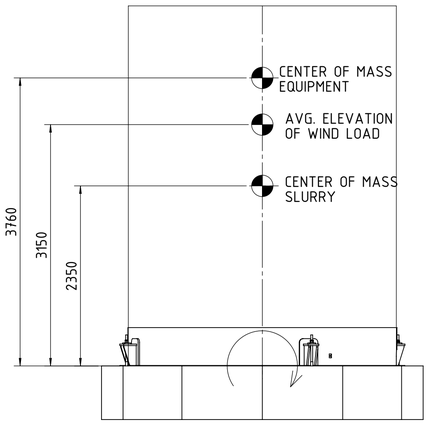
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

ATTACHMENT-1 : LOAD DATA DRAWINGS

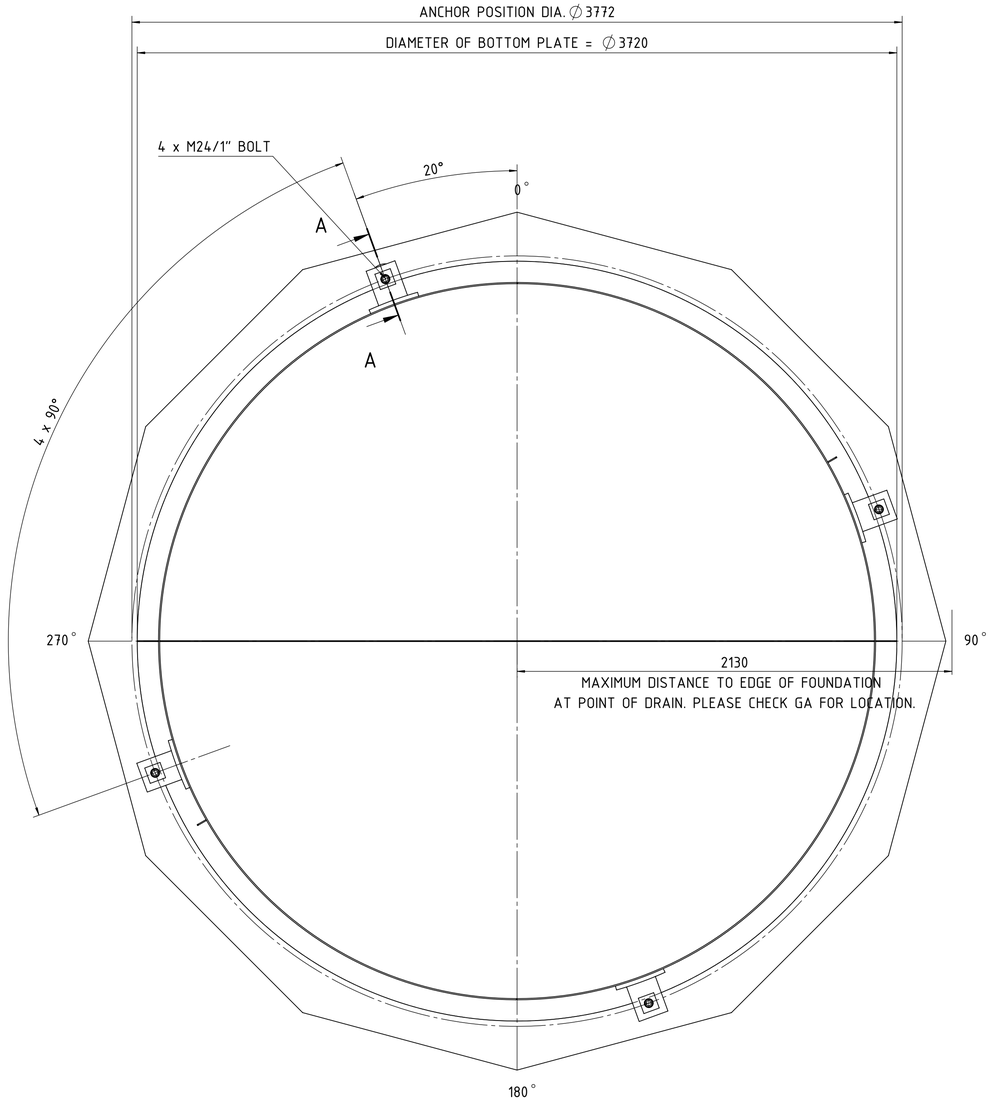
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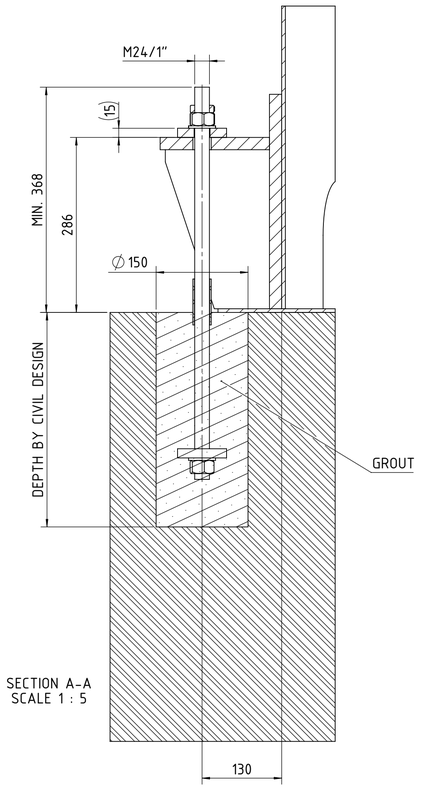
DRAWING NUMBER: MD-216-3610-EG-ME-PLN-1001
REV: D01

LOADING NOTES:	
1.	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2.	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
3.	REPORTED SEISMIC LOADS ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC 301 - 2007. HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNERACCORDING SBC 301 - 2007.
4.	REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC 301 - 2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE SHELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT



SCALE 1 : 40





SECTION A-A
SCALE 1 : 5

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. ground elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

POX CONDITIONING TANK	
DEAD WEIGHT	7400 kg
OPERATIONAL WEIGHT OF SLURRY	52900 kg
MAXIMUM WEIGHT OF SLURRY	59200 kg
EXTERNAL PLATFORM LOAD	1760 kg
NORMAL OPERATING SPEED	62,5 rpm
WIND LOAD	17 kN
SEISMIC LOAD	26,1 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	120 kNm

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3610-EG-ME-GAD-1009...1010	GENERAL ARRANGEMENT								
MD-216-3610-EG-ME-DAT-1005	EQUIPMENT DATASHEET	D01	22.11.2023	AS BUILT	VP	TM	JR	GR	
MD-216-3610-EG-PR-PID-1001...1002	P&I DIAGRAM	C01	03.03.2020	APPROVED FOR CONSTRUCTION	VP	TM	JR	GR	
MD-216-3610-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	15.01.2020	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company شركة المعادن العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-D-02920-001 — REVISION: 3
OUTOTEC DRAWING NUMBER: 900707 GAU01 HL12 MLD15 36101 — DOCUMENT ID: OU601266971 — REVISION: 3

DRAWN	vilpoy	27.02.2020
CHECKED	tuoman	03.03.2020
ENGINEER	juhris	03.03.2020
APPROVED	GeoRie	03.03.2020
EMPLOYER		
SCALE	1:15	A1

MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
POX CONDITIONING, LEACHING AND ADSORPTION
3610-REA-001...002 POX CONDITIONING REACTOR

DRAWING NUMBER: MD-216-3610-EG-ME-PLN-1001 — REV: D01

MA'ADEN معادن Saudi Arabian Mining Company شركة التعدين العربية السعودية		Outotec LARSEN & TOUBRO
MD-216-3610-EG-CV-CAL-1001	Mansourah Massarah Gold Project	
900707_GAU01_HL12_CED01_ 36103_OU100680999_R2	DESIGN CALCULATION POX CONDITIONING, LEACHING & ADSORPTION FOUNDATION FOR REA – 001 & 002	Date: 22.11.2023
O19045_450_C_K_24632_001_R2	AS BUILT	Rev: D01

ATTACHMENT-2 : COMMENT RESOLUTION LOG SHEET/ COMPLIANCE REPORT

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Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.:** 216

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-002345, 5 May , 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 5 May 2020	Comment by	EPC Response DATE: 8 May 2020	Status	Remarks
			General Comments (cc)				
MD-216-3610-EG-CV-CAL-1001 -A01	3610	1	Please notice, if E has been obtained from API 650, then you should use 1E instead of 0,7E.	Mr. Doron Golan BECHTEL	Tank loads are provided by OT calculated as per SBC-301,2007. Load combinations are as per same code i.e. SBC-301, 2007 and hence are correct. The matter was in detail explained to PMC during OT & L&T meeting at Kolkata during Feb 2020 when OT clarified that all tank loads are calculated on basis of SBC-301, 2007. Similar comments were made by PMC in earlier doc # MD-216-3610-EG-CV-CAL-1304-A01 which was clarified later and finally approved by PMC (vide transmittal no. BECHTEL-WTRAN-002211 dated 24.4.20) with load factor 0.7E.	Closed	
MD-216-3610-EG-CV-CAL-1001 -A01	3610	2	Please notice: if E has been obtained from API 650, you should use 1,6E instead of E	Mr. Doron Golan BECHTEL	Tank loads are provided by OT calculated as per SBC-301,2007. Load combinations are as per same code i.e. SBC-301, 2007 and hence are correct. The matter was in detail explained to PMC during OT & L&T meeting at Kolkata on Feb 2020 when OT clarified that all tank loads are calculated on basis of SBC-301, 2007. Similar comments were made by PMC in earlier doc # MD-216-3610-EG-CV-CAL-1304-A01 which was clarified later and finally approved by PMC (vide transmittal no. BECHTEL-WTRAN-002211 dated 24.4.20) with load factor 1.0E.	Closed	